

A Two-step Anomaly Detection Based Method for PU Classification in Imbalanced Data Sets

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Abstract

Several machine learning applications, including genetics and fraud detection, suffer from incomplete label information. In such applications, a classifier can only train from positive and unlabeled (PU) examples in which the unlabeled data consist of both positive and negative examples. Despite a substantial presence of PU learning in the literature, few works have considered a class imbalance setting. Hence, we propose a novel two-step method that exploits anomaly detection to identify hidden positives within the unlabeled data. Our method allows the end-user to choose the anomaly detector depending on preference or domain knowledge. Moreover, we introduce Nearest-Neighbor Isolation Forest (NNIF), a novel semi-supervised anomaly detector based on the Isolation Forest. In contrast to unsupervised anomaly detectors, NNIF can utilize all available label information. Empirical analysis shows that our method generally outperforms, using NNIF as the anomaly detector, state-of-the-art PU learning methods for imbalanced data sets under different labeling mechanisms. Further experiments suggest that our two-step method shows strong robustness to wrong class prior estimates.

Keywords: PU Learning, Weak Supervision, Anomaly Detection, Imbalanced Classification

1 Introduction

Despite the ubiquity of supervised learning in practice, numerous machine learning applications suffer from inaccurate or incomplete label information, e.g. in fraud detection [1], genetics [2–4], and medical diagnosis [5]. As a consequence, there has been a growing interest in the literature to address the limitations of strongly supervised learning [6–8]. In this work, we consider the problem of imperfect label information in the positive (minority) class. This particular setting relates to PU learning [9–11] and one-sided label noise [12]. Positive and unlabeled data or PU learning assumes the labeled examples are positive, but the unlabeled examples can belong to either the positive or negative class. Thus, PU learning is a variant of semi-supervised learning in which negative examples are not available for model training [13]. In contrast, one-sided label noise reflects the problem of inaccurate label information because some examples of one class are incorrectly labeled. In this work, we consider PU data and one-sided label noise as practically equivalent. For a more detailed discussion on the similarities and differences between the two areas, we refer the reader to [7].

PU learning has increasingly gained popularity in recent years, leading to the development of several methods in this area [10, 14–16]. Most of these studies assume that the labeled examples are selected completely at random (SCAR) from the positive examples. This assumption simplifies the problem significantly allowing for PU learning to be addressed using classical binary classification with minor corrections. However, many applications involve a biased subset of labeled examples and are hence not selected completely at random. For instance, in recommendation systems, users often rate the movies they like but are uninterested in rating the movies they dislike. Moreover, most previous works have not considered class imbalanced settings to evaluate the performance of the proposed methods. The lack of empirical comparison of PU learning methods in imbalanced data sets results in an insufficient provisioning of tangible guidance to practitioners in areas such as medical diagnosis [5], and fraud detection [17].

Our contribution to the literature is two-fold. First, we propose a novel two-step method, based on anomaly detection, which is suitable for dealing with PU classification in imbalanced data sets. Second, by comparing state-of-the-art PU learning methods using 16 publicly available imbalanced data sets, we address the lack of empirical comparison in the literature. Our method exploits anomaly detection to identify outliers as positives within the unlabeled data. Then, the outliers are treated according to the end-user’s choice whereas the remainder of the unlabeled examples are considered negative. This method is close to other two-step techniques within the PU literature because it first identifies reliable or unreliable negatives, and secondly, learns a regular supervised model from the treated data. However, we adapt the Isolation Forest (IF) technique [18] to the specific task of finding anomalies with limited known positive examples. As such, we propose the Nearest-Neighbor Isolation Forest (NNIF), which exploits the limited information of the known positive

examples to improve the anomaly scores. We show that our proposed two-step method, with NNIF as the anomaly detector, generally outperforms other state-of-the-art PU learning methods under different experimental settings.

The remainder of the paper is organized as follows. Section 2 provides an overview of the relevant literature. In Section 3, we present our two-step method and NNIF in more detail. Section 4 discusses the relevant aspects of the experimental setup. Section 5 analyses the experimental results. Finally, Section 6 offers conclusions and general advice to the practitioner.

2 Preliminaries

2.1 PU Learning

PU learning is a variant of the semi-supervised setting in which there only exist positive and unlabeled examples. A labeling mechanism, a key concept in PU learning, defines which labeled positive examples are observed. Different assumptions can be made regarding the observation of labeled positive examples or labeling mechanism. Most PU learning methods are built on the selected completely at random (SCAR) assumption. The SCAR assumption states that the positive labeled examples are a randomly selected subset of the set of positives. SCAR implies that supervised techniques can be used for PU learning because the ranking order of predictions is preserved as if the true label was known [10]. The selected at random (SAR) [19] assumption is a more realistic assumption regarding the labeling mechanism. As for SAR, the probability of a positive observation being labeled depends on the features or attributes. The latter assumption is much more sensible for common PU learning applications, including recommendation systems, medical diagnosis, and fraud detection. One specific type of SAR is probabilistic gap (PG), which assumes that the positive examples that resemble negatives more, are less likely to be labeled [16]. In previous studies, the SAR assumption is induced by using a propensity score [19] or a conditional probability [16] to bias the labeled examples. In our research, we compare the different methods under both the SCAR and SAR assumptions.

Most methods in the literature on PU learning can be divided into three categories: two-step techniques, biased learning, and class-prior based methods [7]. The two-step techniques first identify the most reliable negatives and then, a model learns from the newly labeled data. A semi-supervised technique can be used to also exploit the remaining unlabeled data [9]. Early two-step techniques come from the text classification literature [15, 20, 21] in which models such as Naïve Bayes (NB) and support vector machines (SVM) are often used. For instance, the classical S-EM [20] first introduces some labeled instances as “spies” to identify unreliable negatives and then applies semi-supervised NB to predict the labels of the unreliable negatives. Other recent works have proposed two-step methods that exploit clustering to identify likely negatives from the unlabeled data [2–4]. If an unlabeled instance is farther from a positive-class cluster, then the instance is more likely to be a reliable

negative. These methods have been particularly successful in gene reconstruction networks, which is often an imbalanced learning task, where only a few gene pairs have been validated by experts. Clustering-based two-step methods might be better suited to accommodate more realistic labeling mechanisms in which the similarity to the positive class is relevant, such as PG. The two-step approach works well as long as separability exists between the negative and positive class [7]. Biased learning techniques consider unlabeled instances as negatives with label noise [5, 14, 22]. Most biased learning techniques put a higher misclassification cost for false positives by either introducing asymmetric penalization [9] or by using bagging [5, 14]. For instance, PU Bagging selects all labeled examples and takes bootstrap samples of the unlabeled examples. One advantage of PU Bagging is that it enables the user to choose any base learner: SVM is used in [14] but other models such as a decision tree can be selected as well. The class-prior based methods utilize the class prior information to either preprocess the data or modify the algorithm [19, 23–25]. Some class-prior based methods particularly modify the loss function under the empirical minimization framework. This has led to the straightforward adaptation of well-known algorithms such as logistic regression [19, 24] and neural networks [26] for PU learning. Imbalanced nnPU is the only class-prior based method that has adapted the framework to deal with class imbalance. Currently, the class-prior techniques are considered as state-of-the-art solutions for PU learning. Although the PU learning literature has developed a plethora of methods, we see a lack of works in the imbalanced setting or with a different labeling mechanism other than SCAR. Several PU learning methods are not suitable to deal with a low frequency of positives, and even a lower number of labeled positive examples. For instance, the two-step techniques rely on the performance of a (semi-)supervised model, which might be sensitive to such an imbalanced setting. Likewise, most class-prior based techniques generally have no built-in adaptation for the imbalanced setting and shares the same issue as other standard classifiers. Our work precisely addresses this gap in the literature, which is reflected in the choice of the benchmark data sets and the proposed technique.

2.2 Learning under Label Noise & Anomaly Detection

A related area of research to PU learning is learning under label noise. In this field, it is assumed that some examples have a class label that does not correspond to its true value: PU learning can be interpreted as one-sided label noise because the positive examples can only be mislabeled [7]. In fact, the biased learning methods in PU learning assume the one-sided label noise setting.

In [8], it is proposed to divide the existing methods into three categories: cleansing, noise robust, and noise tolerant methods. The cleansing methods require a sensible heuristic that filters the label noise into a cleansed data for a supervised model [27]. Some cleansing methods rely on ad-hoc measures from the model predictions [28, 29] or measures of model influence [30]. The robust noise methods provide partial insensitivity to label noise by applying a robust

loss function [31] or regularization [32]. The robust noise methods are more appropriate in case of limited presence of label noise. Noise tolerant methods aim to explicitly model the label noise, but generally come with stronger assumptions [33–35].

Arguably, the cleansing approach provides the most practical solution for dealing with label noise [8]. It can, however, lead to pitfalls such as over-cleansing [36], which so far has not been widely explored. Despite several works focusing on the cleansing approach, few studies have exploited outlier detection for dealing with label noise [37, 38]. Outlier detection might help particularly in an imbalanced setting. As our work is particularly interested in the imbalanced PU classification, outlier detectors are utilized to cleanse the label noise before the supervised modeling.

The evaluation of unsupervised outlier detectors is a challenge for researchers as several aspects must be assessed in order to identify the weaknesses and strengths of the methods [39]. In empirical studies [40, 41], IF [18, 42], on average, has performed better than other baselines whereas Local Outlier Factor (LOF) [43] and Angle-based Outlier Detection (ABOD) [44] can better detect clustered anomalies. The success of IF has led to the proposal of several variants that adapt the algorithm to a specific goal or extend its capability [1, 45, 46]. Another important aspect is the performance stability of the outlier detector. Given the unsupervised setting, we never know the best parameter values so we should compare the outlier detectors with “reasonable” values or, even better, a range of common values [47]. Motivated by these empirical results, our work develops a tailored IF-based technique for the detection of hidden positives (PU learning) or mislabeled negatives (one-sided label noise) in imbalanced data sets.

3 Methodology

3.1 Two-Step Anomaly based Method

In order to fill the gap in the literature, we propose a two-step anomaly-based method suitable for imbalanced data sets ¹. As in other works in the PU learning literature, we consider the unlabeled data as negative and the hidden positives as mislabeled negatives. Our method focuses on identifying unreliable negatives within the observed negatives. We can use any anomaly detection technique as long as it provides a score rather than a binary outcome. As discussed in the previous section, potential anomaly detectors include LOF, IF and ABOD. Figure 1 provides an overview of our method, which operates in three steps. Nonetheless, because the third step is a rather standardized application of resampling and training the model, our contribution is mainly in the first two steps, hence the reference to our methodology as a two-step approach.

¹A software implementation of our two-step method is available at https://github.com/CarlosOrtegaV/PU_AnomalyDetection

Step 1 identifies unreliable negatives within the data. An anomaly detector is trained to score the negative examples. We do not use the labeled positives to train the anomaly detector to avoid the masking effect which leads to more false negatives [18, 47]. However, an exception is our newly proposed NNIF method, which exploits the positive labeled instances to modify the standard IF score. The end-user sets a treatment ratio value to select the number of top anomalous instances within the observed negative data. For instance, if the treatment ratio is 5%, then the top 5% most anomalous instances are selected. The optimal treatment ratio reflects the proportion of positives within the observed negative data; under perfect anomaly identification, the optimal treatment ratio retrieves all hidden positives (i.e. mislabeled negatives). The optimal treatment ratio can be estimated from domain knowledge, or alternatively from a small labeled data subset [48], or directly from PU data [10, 49, 50]. The optimal treatment is related to the class prior because it can be calculated as follows: $\frac{\alpha - P(l=1)}{P(l=0)}$, where α is the class prior, $P(l=0)$ is the proportion of seemingly negatives, and $P(l=1)$ represents the proportion of known positives.

Step 2 applies a treatment to the selected examples. Together with the treatment ratio, the end-user opts for a treatment option. We consider two options: either removing the selected instances from the training data set, or relabeling them into positive.

Step 3 applies a resampling method to the treated data set and trains a supervised model. Given the issue of class imbalance, we recommend to choose a resampling technique to reach a balanced class ratio, which have been found to improve classification performance [51, 52]; other works have shown the usefulness of sampling methods in churn prediction [53] and fraud detection [54]. Random under-sampling of the majority class and random over-sampling of the minority class are well-known basic sampling methods [55]. More advanced resampling methods have been proposed in the literature [56, 57]. The end-user has the flexibility to choose a resampling method of preference.

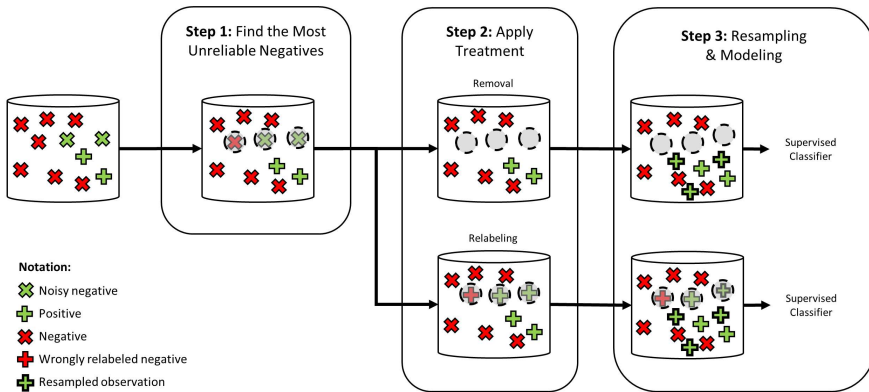


Fig. 1 Diagram illustrating the steps of our approach.

3.2 Nearest-Neighbors Isolation Forest

The goal of the anomaly detector (step 1 of our method) is the identification of mislabeled negatives or noisy negatives. Most well-known outlier detectors are unsupervised so they do not exploit any label information. However, PU classification is a semi-supervised setting so unsupervised techniques fail to use all available information. This issue motivates our proposed IF-based semi-supervised anomaly detector called Nearest-Neighbors Isolation Forest (NNIF). Unlike IF, NNIF incorporates the neighborhood information of the positive labeled data into the anomaly score. Exploiting the neighborhood information to modify the anomaly score of the IF relates to the smoothness assumption. This assumption, also found in other PU learning two-step methods, states that similar instances have a similar classifier output (e.g. conditional probability). In other words, an mislabeled negative with positives in the neighborhood will more likely belong to the positive class than another without close by known positives.

As NNIF extends IF, they both share the same training process. NNIF requires, however, to train a k-nearest neighbor classifier as shown in Algorithm 1. As a consequence, NNIF needs the number of neighbors k as an extra hyper-parameter. In both NNIF and IF, isolation trees (iTree) are constructed as described in Algorithm 1. An iTree is a proper binary tree that recursively splits the feature space into smaller axis-aligned areas. Each iTree uses a subsample X' that is randomly selected without replacement from X so that $X' \subset X$. The splits are chosen randomly across features and values. As discussed in [18, 42], a small subsample size ψ allows IF to be more computational efficient. Moreover, it is suggested that $t = 100$ is enough to provide a stable average path length. At the end of the training process, we have a collection of iTrees and a fitted k-nearest neighbors classifier that can provide the anomaly scores.

Algorithm 1: NN-iForest(X, Y, t, k, ψ, l)

Input: X - training data, Y - labels, t - number of trees, k - number of neighbors, ψ - subsampling size, l - height limit

Output: Forest - set of of t iTrees, $\hat{K}NN$ fitted k-nearest neighbors classifier

```

1 Initialize Forest
2 for  $i \leftarrow 1$  to  $t$  do
3    $X' \leftarrow \text{RandomSample}(X, \psi)$ 
4   Forest  $\leftarrow$  Forest  $\cup$   $iTree(X', l)$ 
5 end
6  $\hat{K}NN \leftarrow \text{FitKNN}(X, Y, k)$ 

```

Algorithm 2 describes in detail the weighted path length, which is needed for the anomaly score in NNIF. For each iTree, a path length $h(x)$ can be

derived as formulated in the standard IF [18, 42]. The path length is calculated by counting the number of edges e from the root node to an external node as an example x traverses through an iTree. If the number of edges e reaches a height limit $hlim$, an adjustment $c(size)$ is added to e for computing $h(x)$. Thus, given an iTree j , the unweighted path length for instance x is computed as follows: $h_j(x) = e + c(size) \in [i, \psi - 1]$. Since an iTree is structurally equivalent to a Binary Search Tree (BST), the adjustment is derived from unsuccessful searches in a BST. The adjustment $c(size)$ estimates the average path length of a random subtree from a number of instances $size$ beyond the tree height limit $hlim$. More formally, the adjustment $c(S)$ is defined as follows:

$$c(S) = \begin{cases} 2H(S-1) - 2(S-1)/S & \text{if } S > 2, \\ 1 & \text{if } S = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where $H(i)$ is the harmonic number and it is estimated by $\ln(i) + e$ (e being Euler's constant). However, NNIF uses the estimated conditional probability of the k -nearest neighbors $\hat{P}(Y = 1 | x)$ to calculate the weighted path length $\tilde{h}(x) = h(x)(1 - \hat{P}(Y = 1 | x))$.

Algorithm 2: WeightedPathLength($x, T, e, K\hat{N}N$)

Input: x - an instance, T - an iTree, e - current path length; to be initialized to zero when first called, $K\hat{N}N$ - fitted k -nearest neighbors classifier

Output: weighted path length of x

```

1  $\hat{P}(Y = 1 | x) \leftarrow K\hat{N}N(x)$ 
2  $W_x = 1 - \hat{P}(Y = 1 | x)$ 
3 if  $T$  is an external node or  $e \geq hlim$  then
4   | return  $W_x(e + c(T.size))$ 
5 end
6  $a \leftarrow T.splitAtt$ 
7 if  $x_a < T.splitValue$  then
8   | return  $PathLength(x, T.left, e + 1)$ 
9 else
10  | return  $PathLength(x, T.right, e + 1)$ 
11 end
```

To determine the anomaly score of an instance x , NNIF averages the weighted path length over the learned T iTrees:

$$E(\tilde{h}(x)) = \frac{\sum_{j=1}^T \tilde{h}_j(x)}{T} = \frac{\sum_{j=1}^T W_x h_j(x)}{T}, \quad (1)$$

where $h_j(x)$ is the path length of x derived from the j^{th} iTree and W_x is the complement of the estimated conditional probability $P(Y = 1 \mid x)$. An anomaly is expected to have a shorter path length than a normal instance when it traverses an iTree from root to the external node. Similarly, an instance is more likely to be an anomaly if more labeled positives are within the k neighborhood. Two instances can have different anomaly scores despite the same path length if one instance is close to labeled positives. The anomaly score $s(x, n)$ is computed as follows:

$$s(x, n) = 2^{\frac{-E(\tilde{h}(x))}{c(n)}}, \quad (2)$$

where $E(\tilde{h}(x))$ is defined as in Equation 1, and $c(n)$ works as a normalization factor to compare with models of different subsample values n . Equation 2 describes that the score lies in the interval $[0, 1]$. The same interpretation as in [18, 42] applies:

- If $E(\tilde{h}(x)) \rightarrow 0, s \rightarrow 1$. If the weighted average path length of instance x is close to zero, the anomaly score will be close to one, thus x is considered as an anomaly.
- If $E(\tilde{h}(x)) \rightarrow n - 1, s \rightarrow 0$. If the weighted average path length of instance x is close to the maximum depth of a binary tree given the subsample n , the anomaly score will be close to zero, thus x is considered as a normal instance.
- If $E(\tilde{h}(x)) \rightarrow c(n), s \rightarrow 0.5$. If the weighted average path length of instance x is close to zero, the anomaly score will be close to 0.5. If all examples have an anomaly score close to 0.5, then it is possible that there are no anomalies in the data.

The new element W_x in Equation 1 enables NNIF to harness the neighborhood information. As W_x lies in the closed interval $[0, 1]$, NNIF works as IF in case there is no available label information. In such case, the $W_x = 1$, thus $\tilde{h}(x) = h(x)$. The more labeled positives are close to instance x , then the W_x will be closer to zero so the anomaly score will approach one. Consequently, NNIF provides an advantage over IF due to this flexibility. Moreover, the masking effect due to clustered anomalies might be diminished because W_x will lower the weighted average path length. As W_x is derived from a fitted k -nearest neighbor model, we have an extra parameter in NNIF. The optimal number of neighbors k will naturally depend on the specific data set or application. However, we decide to set the number of neighbors $k = 5$ in line with scikit-learn.

3.2.1 Comparison Between Anomaly Detectors In Half-Moons Data Set

As an illustrative example, we consider the popular “half-moons” data set that consists of two-dimensional points generated from two interleaved half circles. Figure 2a shows a class imbalanced variant of the data set with a 5% positive ratio: the upper half-moon corresponds to the negative class whereas the lower half-moon corresponds with the positive class. Moreover, half of

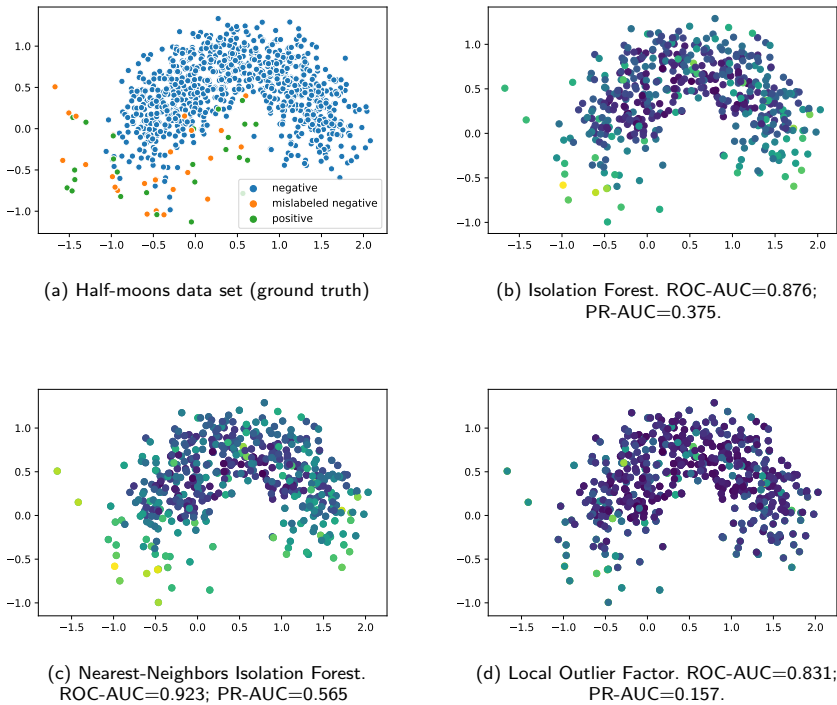


Fig. 2 Comparison of anomaly detectors to find mislabeled negatives on a test set (only including observed negatives). The higher the anomaly score, the lighter the color. The ROC-AUC and PR-AUC are reported in each anomaly detector on an independent test set.

the positives are converted to noisy negatives to induce the label noise. Both IF and LOF are trained using only the negatives, disregarding the available positive examples, whereas NNIF uses all training examples. The anomaly detectors aim to identify the hidden positives among the observed negatives. As we assess the anomaly detectors, only the negatives and the hidden positives are considered for the evaluation. Two metrics are used: the area under the Receiver Operating Characteristic curve (ROC-AUC) and the area under the Precision-Recall curve (PR-AUC). We observe that IF identifies most hidden positives in the lower half-moon. However, some true negatives receive a high anomaly score, which leads to more false positives. LOF retrieves less hidden positives and produces more false positives which correspond to a lower ROC-AUC and PR-AUC. In contrast, NNIF exploits the available information on the known positives to provide a better performance in both evaluation metrics. NNIF especially improves over IF and LOF in the PR-AUC.

4 Experimental Setup

An experimental study is conducted to assess the performance of our two-step method. An additional goal of our empirical analysis is the comparison of the PU methods on imbalanced data sets in two labeling mechanisms (i.e., SCAR and SAR). The experiments are carried out on public data sets for binary classification that reflect the imbalanced setting. Figure 3 shows an overview of the experimental design. As the data sets are not originally PU data, the experimental setup follows the interpretation of PU data as in one-sided label noise [7]: the negative instances are interpreted as unlabeled data so that the positive unlabeled examples are noisy negatives (i.e. mislabeled instances). Another difference compared to most previous studies [10, 14, 22] is that we include both the selected completely at random (SCAR) as well as the selected at random (SAR) assumption to evaluate the methods.

Our two-step method is compared by benchmarking against both a variety of well-known and state-of-the-art PU techniques as well as XGBoost as a naïve solution. For our method, we explore the combinations of two anomaly detectors and two treatments (IF and NNIF) with two treatment options (removal and relabeling). An overview of the configurations of our two-step approach is provided in the top part of Table 2.

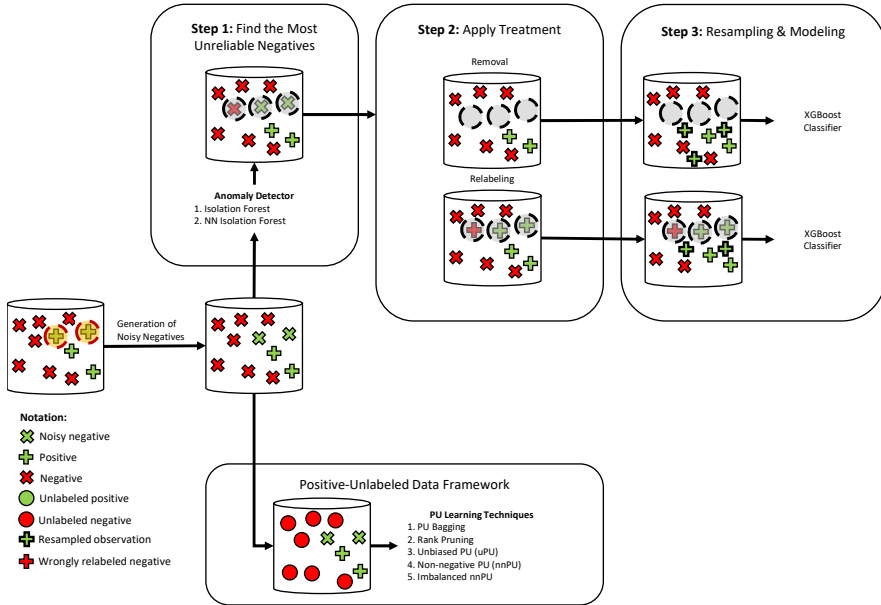


Fig. 3 Diagram illustrating the experimental setup.

4.1 Benchmark Data Sets

We conduct experiments on 16 data sets that cover different domains such as medical diagnosis, bank marketing, fraud and image detection. The data sets are well-known in the anomaly detection and imbalanced classification literature [46, 58, 59]. The data sets are summarized in Table 1. Nine of the data sets can be found in ODDS [60], three in OpenML [61], two in KEEL [62], the fraud insurance data set comes from Oracle [63], and the bank marketing data set comes from the UCI repository [64]. Note that we used a random subset of four of the data sets to limit the computation time of the algorithms: we randomly sample 5000 observations in forest cover, mammography, shuttle, and satellite.

In Table 1, “Pos. Class Ratio” (positive class ratio) shows the class imbalance in the data sets. We opt to artificially create PU data by flipping positive instances into noisy negatives (i.e., mislabeled positives) because there is a lack of available real PU data sets or data sets with one-sided label noise [7, 8, 65, 66]. “Pol. Ratio” (pollution ratio) is the proportion of mislabeled positives within the seemingly negative instances; Pollution ratio can be also understood as the optimal treatment ratio. Different pollution ratios can be obtained depending on the flip ratio which is the proportion of positive examples that are flipped to noisy negative. We consider three values for the flip ratio: 25%, 50%, and 75%. Whereas the flip ratio determines the number of flipped positives, the labeling mechanism defines which of the positive examples are flipped. In the SCAR assumption, positive instances are randomly flipped regardless of the features. In the SAR assumption, the flipped examples are a biased subset of the positive class, where the bias depends on the features. We propose a specific implementation of the SAR assumption that uses the average euclidean distance to bias the label mechanism: positive examples that are closer on average to the negative class, are more likely to be flipped. The distance-based labeling mechanism shares some similarity to the probabilistic gap assumption [16], however the latter relies on an SVM classifier to determine the probability of being flipped. To the best of our knowledge, only two studies have previously considered the SAR assumption in their empirical evaluation [16, 19]. For each data set, we perform 20 repetitions of a holdout validation that splits the data into a training set (70%) and test set (30%). In total, there are 1920 settings (16 datasets $\times$ 20 repetitions $\times$ 2 labeling mechanisms $\times$ 3 flip ratios). In all the experiments, we ensure experimental reproducibility through the random state parameter; in each repetition, a different random state is utilized. This parameter determines the randomness in both experimental procedures (e.g., train and test sampling) and algorithms (e.g., XGBoost’s feature sampling).

Table 1 Summary of data sets

Data set	Obs.	Features	Pos. Class Ratio	Pol. Ratio at 25%	Pol. Ratio at 50%	Pol. Ratio at 75%
cardio	1831	21	9.60%	2.59%	5.04%	7.38%
thyroid	3772	7	2.50%	0.69%	1.27%	1.89%
climate	540	20	8.52%	2.28%	4.45%	6.53%
forest cover	5000	10	1.00%	0.25%	0.50%	0.75%
seismic	2584	11	6.57%	1.73%	3.40%	5.01%
mammography	5000	6	2.58%	0.66%	1.31%	1.95%
shuttle	5000	9	6.40%	1.68%	3.31%	4.88%
letter	1600	32	6.25%	1.64%	3.23%	4.76%
yeast	1004	8	9.86%	2.66%	5.19%	7.58%
poker 8-9 vs 5	2075	10	1.20%	0.30%	0.61%	0.91%
wine quality red 4	1599	11	3.31%	0.85%	1.69%	2.51%
pie chart 2	745	36	2.15%	0.55%	1.09%	1.62%
pizza cutter 1	661	37	7.87%	2.09%	4.09%	6.02%
satellite	5000	36	1.48%	0.37%	0.75%	1.11%
fraud insurance	15420	30	5.98%	1.57%	3.08%	4.55%
bank marketing	45211	16	11.69%	3.20%	6.21%	9.03%

4.2 Baseline Methods & Parameter Configuration

We compare our two-step method with eight state-of-the-art PU learning techniques: Spy-EM [20], Elkan-Noto’s method [10], Weighted PU Logistic Regression [9], PU Bagging [14], Rank Pruning [67], Unbiased PU [24], Non-negative PU [26], and Imbalanced Non-negative PU [23]. We also include a naïve baseline which is a XGBoost [68] with ADASYN on the PU data.

Table 2 summarizes the hyper-parameter configuration of the methods in the experimental setup. PU learning methods in our experimental setup require hyper-parameters that need to be specified by the end-user. However, the authors have suggested values for most of the hyper-parameters that have shown good results in previous experiments. In contrast to PU learning methods, expecting to find optimal hyper-parameters for anomaly detectors is not realistic because of the unsupervised setting. At best, an end-user can select hyper-parameters based on a range of reasonable values [47]. Thus, we opt for using default values in `scikit-learn` for the hyper-parameters of IF and NNIF. The hyper-parameters of semi-supervised techniques are either chosen by previous works [69] or default values of the software implementation. One important hyper-parameter in our method is the treatment ratio which reflects, given a perfect precision of the anomaly detector, the proportion of mislabeled positive examples within the observed negative data. This contrasts with the class prior-based methods that require an estimation of the proportion of positives in the data. The focus on the pollution ratio allows our method to deal with both SCAR and SAR assumptions. Therefore, we set the treatment ratio to the pollution ratio at each flip ratio. This is similar to other works in the PU literature in which the true class prior is given [23, 24]. This follows a scenario that corresponds to selecting a reasonable value based on the domain knowledge.

Table 2 Experimental hyper-parameters of the Two-Step Anomaly-based Method and PU learning techniques

Configuration	Setting
Anomaly Detectors	Nearest-Neighbors Isolation Forest (neighbors = 5, other default hyper-parameters as in <code>scikit-learn</code>) Isolation Forest (default hyper-parameters as in <code>scikit-learn</code>)
Treatment	None Relabeling Removal
Treatment Ratio	Set to the pollution ratio in each data set
Resampler	ADASYN [57] (neighbors = 5, sampling strategy = balanced ratio)
Classifier	XGBoost [68] (default hyper-parameters)
PU techniques	uPU (classifier - XGBoost, default hyper-parameters as suggested by [24]) nnPU (classifier - XGBoost, default hyper-parameters as suggested by [26]) imbalanced nnPU (classifier - XGBoost, default hyper-parameters as suggested by [23]) PU Bagging (base learner = SVM with RBF kernel as in [14], learners = 100, k = positives) Rank Pruning (default hyper-parameters as suggested by [67]) Spy-EM (default hyper-parameters as suggested by [20]) PU Weighted Logistic Regression (weights as in [22])

5 Results & Discussion

5.1 Analysis on Aggregated Results

The performance of each method is measured across 16 imbalanced data sets and different flip ratios and labeling mechanisms. We separately examine the average performance metrics under SCAR and SAR. Firstly, different combinations of our method are compared to select the best performing model as shown in Table 3 and Table 4. Once the best performing model is identified, we contrast the performance of our method with other techniques in a global comparison. We apply hypothesis testing to further validate the empirical findings in the global comparison. The Iman-Davenport test [70] uses the average rank of the evaluation metrics to decide whether the null hypothesis of all methods performing equally is rejected. Furthermore, we apply the Holm's post-hoc test for pairwise comparison of the best performing model with all other techniques in order to gain a more fine-grained perspective on performance differences.

In the comparison between different combinations of the anomaly-based two-step method (Table 3 and Table 4), we find that NNIF generally outperforms IF given the same treatment in both SCAR and SAR: for instance, NNIF and removal treatment generally outperforms IF and removal treatment. This result suggests that incorporating the label information of the neighborhood into the anomaly score leads to a performance improvement. Another finding is that the choice of treatment has an impact on the performance. Removal outperforms relabeling in terms of the PR-AUC, regardless of the labeling mechanism. Relabeling tends to outperform removal in terms of the ROC-AUC when NNIF is used as the anomaly detector; however, relabeling with IF as the anomaly detector is generally outperformed by removal. This result indicates that the performance of relabeling is more sensitive to false positives than removal: relabeling true negatives tends to be more harmful than removing them. This problem is less noticeable when NNIF is used because

of the improved anomaly score. Depending on the main metric, the best performing method can either be NNIF with removal or NNIF with relabeling. In imbalanced classification, however, the end-user might be more interested in precision because of practical reasons: in fraud detection, the investigators process a limited amount of cases so that precision is often the main objective [71]. Accordingly, NNIF with removal is selected as the best performing model of our two-step method.

Table 3 Evaluation metrics, expressed as percentage, under the SCAR assumption across different flip ratios

Model Flip Ratio:	ROC-AUC (SD)			PR-AUC (SD)		
	25%	50%	75%	25%	50%	75%
IF Relabeling	86.3 $\pm$ 2.80	84.0 $\pm$ 3.55	80.2 $\pm$ 4.37	48.4 $\pm$ 5.16	41.3 $\pm$ 5.91	34.3 $\pm$ 5.23
IF Removal	87.2 $\pm$ 2.84	85.2 $\pm$ 3.78	80.5 $\pm$ 4.90	53.5 $\pm$ 5.66	49.8 $\pm$ 6.37	42.6 $\pm$ 6.80
NNIF Relabeling	87.3 $\pm$ 2.83	85.3 $\pm$ 3.55	82.2 $\pm$ 4.20	52.6 $\pm$ 5.31	48.2 $\pm$ 5.81	39.8 $\pm$ 5.49
NNIF Removal	87.3 $\pm$ 2.82	85.7 $\pm$ 3.38	81.8 $\pm$ 5.24	53.7 $\pm$ 5.38	50.9 $\pm$ 5.95	46.1 $\pm$ 6.50

Best performing model is in **bold and underlined** in each column.

Table 4 Evaluation metrics, expressed as percentage, under the distance-based labeling mechanism (SAR assumption) across different flip ratios

Model Flip Ratio:	ROC-AUC (SD)			PR-AUC (SD)		
	25%	50%	75%	25%	50%	75%
IF Relabeling	86.1 $\pm$ 3.20	83.6 $\pm$ 3.74	80.1 $\pm$ 4.17	48.0 $\pm$ 5.18	40.7 $\pm$ 5.97	34.0 $\pm$ 5.48
IF Removal	86.9 $\pm$ 2.93	85.0 $\pm$ 3.78	80.6 $\pm$ 5.33	52.7 $\pm$ 5.20	49.2 $\pm$ 5.76	44.4 $\pm$ 6.80
NNIF Relabeling	87.4 $\pm$ 2.81	85.6 $\pm$ 3.21	82.1 $\pm$ 4.28	53.3 $\pm$ 5.19	48.7 $\pm$ 5.62	39.5 $\pm$ 5.84
NNIF Removal	87.2 $\pm$ 3.08	85.2 $\pm$ 3.84	80.3 $\pm$ 4.91	53.7 $\pm$ 6.04	50.4 $\pm$ 6.00	43.4 $\pm$ 6.64

Best performing model is in **bold and underlined** in each column.

In the global comparison, the average and standard deviation of the ROC-AUC and PR-AUC are shown in Table 5 and Table 6. Notice that the setting without label noise is included as a reference of the ideal performance given XGBoost as the classifier. The Iman-Davenport test [70] rejects the null hypothesis that all methods perform the same ($p < 0.01$). We find that NNIF with removal treatment achieves the best performance in most of the experimental settings. Nevertheless, PU Bagging performs almost the same in ROC-AUC at a flip ratio of 75%. At a flip ratio of 25%, our method does not significantly outperform state-of-the-art techniques such as nnPU [26] and Imbalanced nnPU [23]. Although all techniques obtain a lower performance with a higher flip ratio, the anomaly-based two-step method is generally better suited for medium and high label noise settings: our method significantly outperforms all baselines in PR-AUC at flip ratios of 50% and 75%.

We can point out several observations from the comparative analysis. At a low flip ratio, XGBoost with ADASYN performs relatively well compared to other state-of-the-art techniques, despite learning from noisy labels. This

Table 5 Evaluation metrics, expressed as percentage, under the SCAR assumption across different flip ratios

Model	Flip Ratio:	ROC-AUC (SD)			PR-AUC (SD)		
		25%	50%	75%	25%	50%	75%
NNIF Removal		87.3 ± 2.82	85.7 ± 3.38	81.8 ± 5.24	53.7 ± 5.38	50.9 ± 5.95	46.1 ± 6.50
IF Removal		87.2 ± 2.84	85.2 ± 3.78	80.5 ± 4.90	53.5 ± 5.66	49.8 ± 6.37	42.6 ± 6.80
SpyEM		77.7 ± 5.25	78.3 ± 5.73	76.5 ± 8.30	32.8 ± 5.46	34.0 ± 6.88	35.8 ± 8.64
Weighted PU Log. Reg.		84.0 ± 3.59	81.9 ± 4.44	79.6 ± 4.78	49.1 ± 5.22	46.8 ± 6.36	42.0 ± 7.09
Elkan-Noto's Method		73.3 ± 7.90	70.6 ± 11.32	62.6 ± 16.45	44.5 ± 6.83	36.9 ± 10.11	25.0 ± 12.98
PU Bagging		83.9 ± 3.25	82.9 ± 3.60	81.7 ± 4.04	41.5 ± 5.11	40.9 ± 5.72	40.0 ± 6.45
Rank Pruning		83.8 ± 3.74	82.2 ± 4.66	79.4 ± 5.20	48.2 ± 5.66	46.3 ± 7.05	41.6 ± 7.27
uPU		85.5 ± 3.58	82.5 ± 5.08	77.3 ± 5.71	50.1 ± 5.99	44.9 ± 6.69	37.4 ± 6.68
nnPU		86.6 ± 3.25	84.4 ± 4.27	81.0 ± 5.34	52.7 ± 5.77	49.3 ± 6.03	43.7 ± 6.83
Imbalanced nnPU		86.5 ± 3.27	84.1 ± 4.43	80.4 ± 5.30	52.7 ± 5.73	49.3 ± 6.29	43.7 ± 6.92
XGBoost		87.0 ± 3.03	84.8 ± 3.92	80.5 ± 5.61	52.9 ± 5.64	48.7 ± 6.04	42.3 ± 7.10
Iman-Davenport test		$p < 0.01$	$p < 0.01$	$p < 0.01$	$p < 0.01$	$p < 0.01$	$p < 0.01$
XGBoost (No Label Noise)		88.4 ± 2.94	88.4 ± 2.94	88.4 ± 2.94	56.0 ± 4.95	56.0 ± 4.95	56.0 ± 4.95

Best performing model is in **bold and underlined** in each column.

Values in **bold** indicate that the best performing model does not outperform the classifier in the model column at the 5% significance level.

result holds regardless of the labeling mechanism. Generally, the best performing PU methods are the techniques based on the unbiased PU learning framework [24]: nnPU and Imbalanced nnPU perform the best, which use the modification proposed by [26] so that overfitting is less severe compared to standard unbiased PU learning. PU Bagging is the second best-performing technique in terms of ROC-AUC at a flip ratio of 75% in both SCAR and SAR. In line with previous works [22, 67], Rank Pruning and PU Weighted Logistic Regression outperform Elkan-Noto and Spy-EM according to both metrics and labeling mechanisms. The overall worst performing technique is Elkan-Noto in terms of ROC-AUC, whereas Spy-EM underperforms in terms of PR-AUC.

We can notice that most of the techniques are not well suited for PU data with class imbalance. Some PU techniques perform worse than naively using XGBoost with ADASYN on the noisy data. Among the techniques, PU Bagging and Imbalanced nnPU are the only techniques that use an algorithmic modification to deal with class imbalance. Despite the algorithmic advantage, NNIF with Removal generally outperforms these techniques with statistical significance. Furthermore, NNIF with Removal is more robust to different kinds of label noise in terms of PR-AUC as the difference in performance between SAR and SCAR is lower compared to PU Bagging and Imbalanced nnPU: most techniques are only suited to the SCAR assumption, which is often unrealistic in real-world applications.

Table 6 Evaluation metrics, expressed as percentage, under the distance-based labeling mechanism (SAR assumption) across different flip ratios

Model	Flip Ratio:	ROC-AUC (SD)			PR-AUC (SD)		
		25%	50%	75%	25%	50%	75%
NNIF Removal		87.2 ± 3.08	85.2 ± 3.84	80.3 ± 4.91	53.7 ± 6.04	50.4 ± 6.00	43.4 ± 6.64
IF Removal		86.9 ± 2.93	85.0 ± 3.78	80.6 ± 5.33	52.7 ± 5.20	49.2 ± 5.76	44.4 ± 6.80
SpyEM		76.2 ± 6.72	73.7 ± 6.98	71.5 ± 8.04	32.8 ± 5.12	31.4 ± 6.62	33.9 ± 7.64
Weighted PU Log. Reg.		83.5 ± 4.30	81.3 ± 4.74	76.9 ± 5.79	48.1 ± 5.50	45.7 ± 6.03	39.3 ± 7.03
Elkan-Noto's Method		73.9 ± 7.92	69.5 ± 13.16	63.3 ± 16.62	43.7 ± 7.46	34.4 ± 11.61	23.2 ± 12.85
PU Bagging		83.8 ± 2.83	82.6 ± 3.42	80.5 ± 4.18	40.8 ± 4.94	39.4 ± 5.27	36.7 ± 5.94
Rank Pruning		83.3 ± 4.31	81.3 ± 4.86	76.3 ± 6.47	47.9 ± 5.33	45.1 ± 6.79	38.4 ± 6.94
uPU		84.9 ± 3.88	82.0 ± 4.84	75.5 ± 6.50	49.5 ± 6.21	43.5 ± 6.40	33.7 ± 6.42
nnPU		86.6 ± 3.19	84.3 ± 3.69	79.3 ± 5.40	52.6 ± 5.46	47.9 ± 6.13	40.8 ± 6.54
Imbalanced nnPU		86.3 ± 3.24	83.9 ± 4.02	79.1 ± 5.31	52.5 ± 5.54	47.9 ± 6.45	41.2 ± 7.11
XGBoost		86.8 ± 2.92	84.4 ± 3.91	78.7 ± 5.23	52.5 ± 5.99	47.5 ± 6.57	39.1 ± 6.94
Iman-Davenport test		$p < 0.01$	$p < 0.01$	$p < 0.01$	$p < 0.01$	$p < 0.01$	$p < 0.01$
XGBoost (No Label Noise)		88.4 ± 2.94	88.4 ± 2.94	88.4 ± 2.94	56.0 ± 4.95	56.0 ± 4.95	56.0 ± 4.95

Best performing model is in **bold and underlined** in each column.

Values in **bold** indicate that the best performing model does not outperform the classifier in the model column at the 5% significance level.

5.2 Analysis on Fraud Insurance & Satellite Data Sets

We present detailed results for two data sets that particularly reflect the scenario in which our two-step method is specialized: severely imbalanced PU classification. The two data sets are fraud insurance and satellite. Notably, fraud detection represents an interesting application because hidden fraudsters (i.e., mislabeled positives) might behave according to the SAR assumption in which hidden fraudsters are closer to the non-fraudster class. Besides NNIF Removal and IF Removal, we select PU Bagging and nnPU given their high performance in either ROC-AUC and PR-AUC in the previous section 5.1. Figure 5 and Figure 4 show the results per data set and labeling mechanism (SCAR or SAR assumption) in boxplots grouped by the flip ratio. As a general pattern, we observe that the PU methods drop in performance with a higher flip ratio and the variability of the evaluation metrics increases. The increased variation in results at a high flip ratio has been reported before as a result of a more severe class imbalance [14].

In the fraud insurance data set (Figure 4), NNIF Removal presents a slight but consistent advantage over IF Removal in both SCAR and SAR assumptions. Despite an underwhelming performance, PU Bagging presents the least variable performance across all experimental settings among the selected methods; at the highest flip ratio, PU Bagging becomes competitive in ROC-AUC as its performance is similar to the other methods and does not require a class prior estimate. The state-of-the-art nnPU performs slightly worse than either NNIF Removal and IF Removal in PR-AUC across all experimental settings; however, for ROC-AUC, nnPU shows a close median performance to our two-step method.

The satellite data set (Figure 5) presents a scenario in which NNIF Removal dominates IF Removal in both evaluation metrics over the experimental settings. Given the same removal treatment, NNIF Removal shows that utilizing NNIF as the anomaly detector improves substantially the two-step method.

In this data set, nnPU outperforms IF Removal but fails to surpass NNIF Removal's performance. Unlike the previous dataset, PU Bagging outperforms nnPU in ROC-AUC for both SCAR and SAR assumptions. NNIF Removal, despite being the best-performing PU method, performs similarly to PU Bagging and nnPU in ROC-AUC over the experimental settings; nonetheless, NNIF Removal still dominates the rest of the PU methods in terms of PR-AUC, particularly at 75% flip ratio.

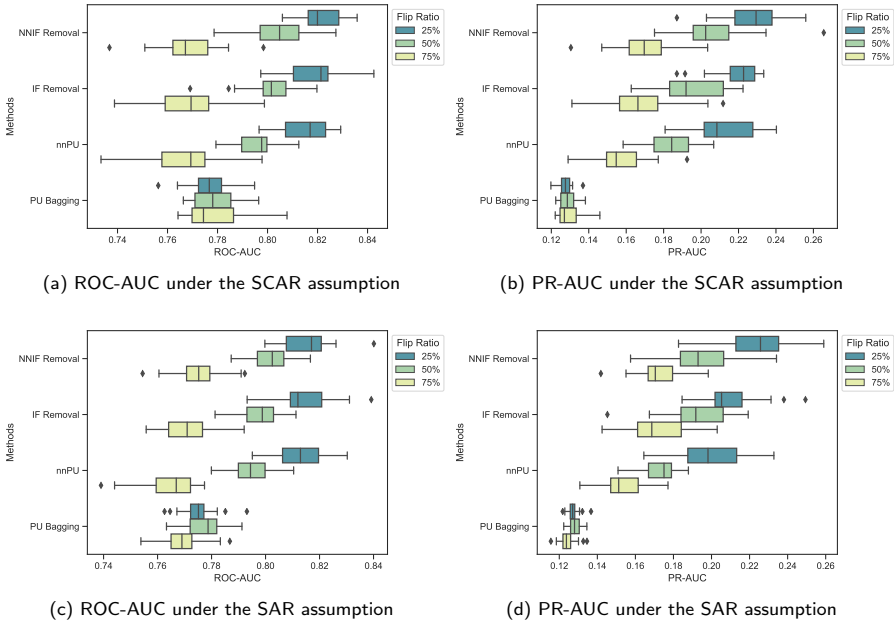


Fig. 4 Comparison of ROC-AUC and PR-AUC scores across different flip ratios in fraud insurance data set

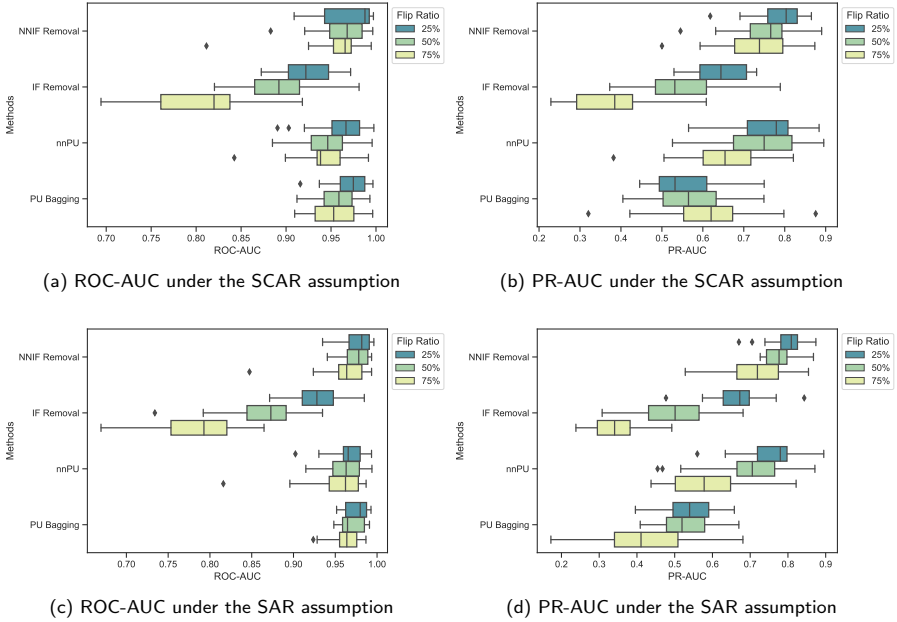


Fig. 5 Comparison of ROC-AUC and PR-AUC scores across different flip ratios in satellite data set

5.3 Sensitivity Analysis on Class Prior Estimate

The accurate estimation of the class prior is essential for most of the state-of-the-art PU methods [7]. Our PU tree-based methods, as other class-prior based methods, In our two-step method, the class prior α can be exploited to calculate the optimal treatment ratio, which is the pollution ratio $P(y = 1 \mid l = 0) = \frac{\alpha - P(l=1)}{P(l=0)}$ as defined in Section 3.1. Several studies have proposed techniques to provide an estimate of the class prior [25, 49, 72] under the SCAR assumption. In this work, we opt for a sensitivity analysis to measure the change in performance when the class prior is wrongly estimated. The sensitivity analysis focuses on the SCAR setting with a 50% flip ratio to represent a moderate scenario in which half of the positives are mislabeled: for instance, it is common to suffer from more than 50% fraudsters being mislabeled in car insurance fraud detection [73, 74]. Similar to subsection 5.1, we select NNIF Removal, IF Removal, and nnPU for the sensitivity analysis; PU Bagging is omitted as the method does not rely on a class prior estimate. We utilize different values of the class prior estimate based on the ground-truth class prior α : the sensitivity analysis includes five values for the class prior estimate $\hat{\alpha} \in \{0.25\alpha, 0.50\alpha, \alpha, 2\alpha, 4\alpha\}$. We perform the analysis per data set with 20 repetitions for each experimental setting.

Figure 6 and Figure 7 present the results of the sensitivity analysis. We can observe that the three PU methods are generally robust to wrong class prior

estimates in ROC-AUC and PR-AUC. The robustness of nnPU in our experiments confirms previous results [26]: due to the modified PU loss function that prevents a negative empirical risk, nnPU offers stronger robustness compared with uPU [24]. In contrast to nnPU, we can observe that overestimation of the class prior can lead to an improvement in ROC-AUC and PR-AUC for NNIF Removal and IF Removal in some data sets (e.g., seismic, pizzacutter1, and yeast). One possible explanation is that the overestimation of the class prior facilitates the classification task: removal treatment further separates the positive and negative distribution in the feature space. Two-step methods can possess an advantage over other PU methods if the class separation is clear enough [7].

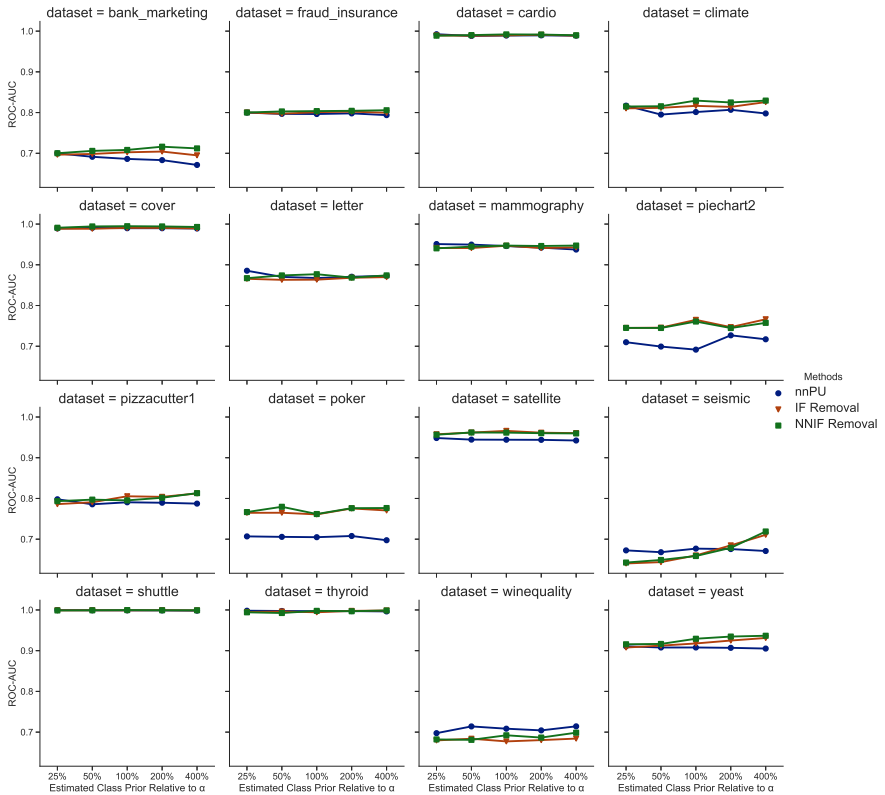


Fig. 6 ROC-AUC per data set at 50% flip ratio under the SCAR assumption across different class prior estimates

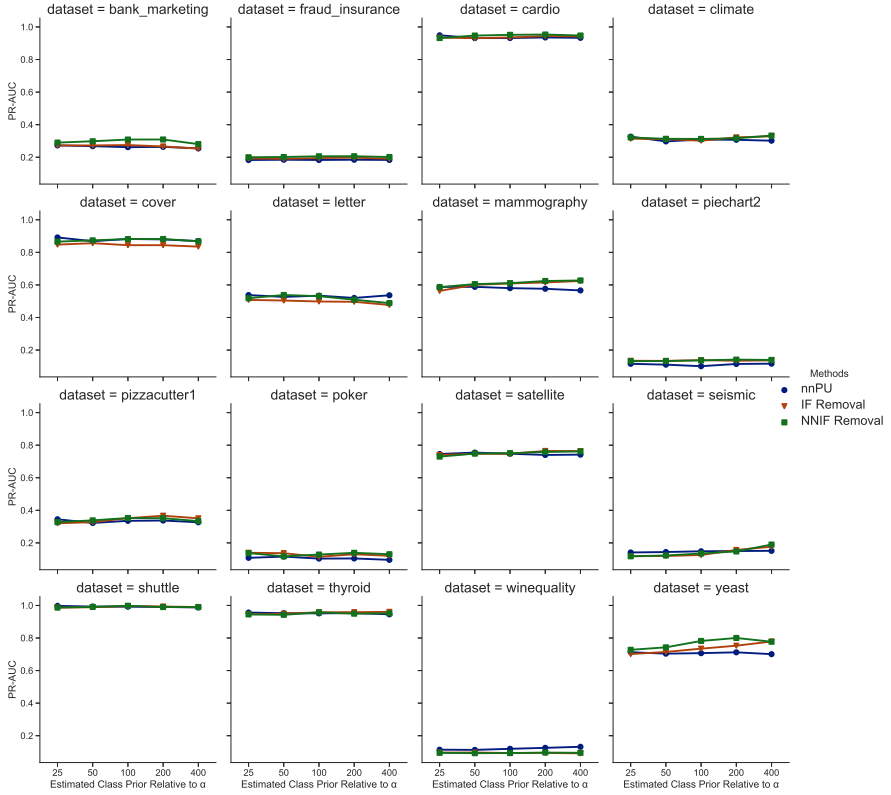


Fig. 7 PR-AUC per data set at 50% flip ratio under the SCAR assumption across different class prior estimates

6 Conclusions & Future Work

Despite the extensive work in PU learning, the application to imbalanced class data sets has not received significant attention in the literature. We address this gap by proposing a novel two-step method that is suitable for imbalanced class PU classification. Moreover, we introduce NNIF, a semi-supervised anomaly detector that incorporates the limited information of well-known positives into the anomaly score. The proposed two-step method allows the end-user to choose the anomaly detector and treatment of preference. The experiments point out that NNIF is the overall best anomaly detector. Furthermore, the best treatment for NNIF is removal as the highest PR-AUC is achieved. We have empirically shown that most PU learning methods are sensitive to the class imbalance setting and a more intricate labeling mechanism than SCAR. The study shows that our two-step method generally outperforms all other techniques under both SAR and SCAR. This includes state-of-the-art techniques such as nnPU and Imbalanced nnPU. The results point out the advantage of our method in several applications, such as fraud detection, where

the SCAR assumption is not valid. The state-of-the-art performance of the proposed method emphasizes the usefulness of anomaly detection in PU learning, which so far has been unexplored in previous studies. Furthermore, we show that our two-step method presents better robustness to underestimation of the class prior than the state-of-the-art nnPU.

For future work, we aim to further validate our new technique with larger data sets in which algorithmic scalability becomes more relevant as well. For instance, NNIF can be optimized for large data sets as it uses k-nearest neighbor classification under the hood. Moreover, we assume that the end-user can provide a reasonable treatment ratio, which might not be possible in some circumstances. Thus, another possibility for future research is to develop a strategy for finding the optimal treatment ratio.

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